

SHANE ROWAND

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

B.S. in Mechanical Engineering — Expected May 2030

- Relevant Coursework: Multivariable Calculus, Linear Algebra, Physics I & II, Introduction to Robotics, Computer Science I, Computer-Aided Design (Siemens NX)

PROJECTS

Axle — Personal AI Assistant & Legged-Robot Platform

Personal Project

- Designed and built a hub-and-spoke, WebSocket-based AI agent system in Python with SQLite-backed session memory, a VS Code extension panel, and a custom HUD-styled web interface
- Implemented Docker-sandboxed command execution, computer-use automation (pyautogui), a Chrome browser bridge, and voice input to give the agent safe, multi-modal control of a desktop environment
- Extending the platform into a hexapod legged robot, architecting a two-layer control system that separates high-level reasoning (LLM) from a low-level locomotion policy
- Training a reinforcement learning locomotion policy (PPO, domain randomization) in MuJoCo/MJX at 50–200 Hz control rates, starting with a single-leg prototype before scaling to the full hexapod
- Modeling leg assemblies in CAD for URDF export into the simulation pipeline

Seatbelt Usage Tracker — PennDOT Competition

Prototype

- Designed a clamp-on device to track seatbelt buckle usage, addressing a road-safety data problem posed by PennDOT
- Modeled the housing in Tinkercad and 3D printed a working prototype to present at the competition

Spotify Barcode Sign

Personal Project

- Fabricated a 3-foot wooden sign in walnut and cherry, precisely replicating the dimensions of a scannable Spotify barcode linking to a custom playlist
- Cast resin inlays and wired an embedded LED strip around the border to backlight the sign, combining woodworking with electrical wiring

TECHNICAL SKILLS

Programming: Python, self-taught software development

CAD: Siemens NX, SolidWorks, Fusion 360, OnShape

Robotics & Simulation: MuJoCo/MJX, reinforcement learning (PPO), URDF, Docker

Platforms: Windows, Git, VS Code

ACTIVITIES & LEADERSHIP

RPI Robotics Club

Troy, NY

Member

2026 – Present

- **Combat Robot:** Designed a 1lb combat robot in OnShape, 3D printed and manufactured all the components, assembled and wired the robot, then tested and competed against other teams after some months of development and production

PAST WORK EXPERIENCE

Weis Markets

Perkasie, PA

Produce Associate

2024 – Present